

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**Department of Computer Science and Engineering****ASSESSMENT TOOL 2 – Case Study Analysis**

Course Code:	CSA65	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO2 – Precision (BL1, BL2)	Assessment:	Assessment Tool 2 – Case Study Analysis
Weightage:	20% of CIA	Total Marks:	100 (1 Question)
CO1 Statement:	<i>Apply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems.(BL1, BL2)</i>		
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Section / Batch:	CSE – AI / 2024	Date:	

Instructions to Students

- Answer any ONE questions.
- Total 100 marks
- Include architecture, explanation, suitable Python/API approach, advantages, limitations and evaluation.
- Time allotted: 30 mins

Code of Conduct

I J KARTHIK / 192472136 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: J KARTHIK

SECTION A – Case Study Analysis

Q20. Case Study 20

Develop an appropriate solution for the given Generative AI/LLM scenario. Your answer should include analysis, justification, suitable model selection (GPT/BERT/RLHF/LoRA/PEFT where applicable), implementation approach, expected outcome, and limitations.

Topic: Hybrid GPT+BERT+RAG system.

1. Problem Analysis

Educational institutions and organizations require an AI-powered assessment system that can answer students' questions accurately based on their syllabus or institutional documents.

Traditional GPT models generate fluent answers but sometimes produce hallucinations (incorrect information). BERT understands the meaning of the user's question very well but cannot generate complete responses. Retrieval-Augmented Generation (RAG) retrieves relevant information from a trusted knowledge base before GPT generates the final answer.

Therefore, combining GPT + BERT + RAG provides both accuracy and natural language generation.

2. Proposed Hybrid Solution

The proposed system consists of three AI components:

BERT

- Understands the user's question
- Extracts semantic meaning
- Generates embeddings
- Improves retrieval accuracy

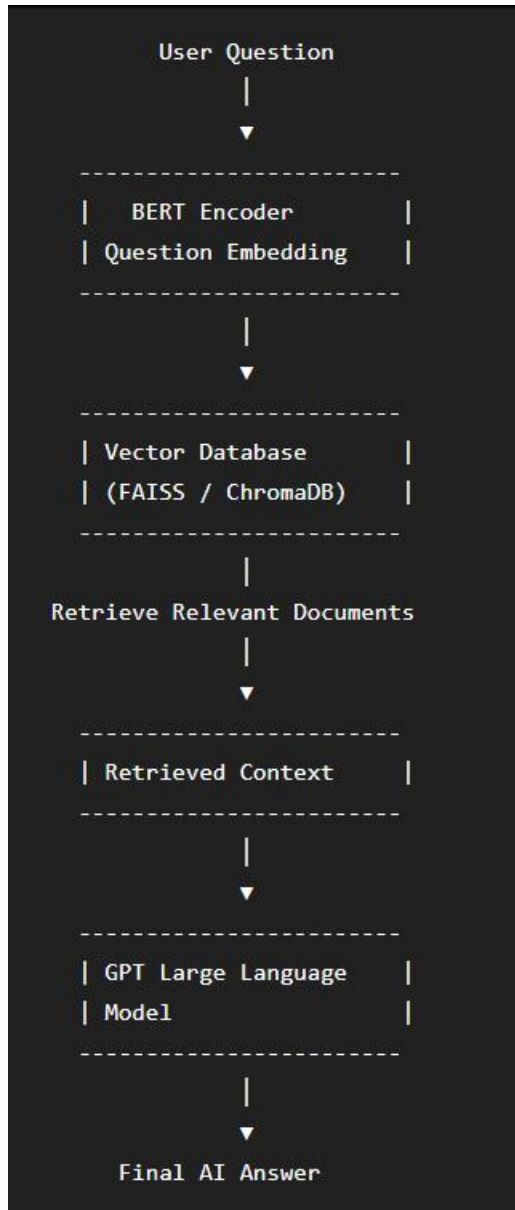
RAG

- Searches relevant documents
- Retrieves syllabus, notes, PDFs, textbooks
- Supplies trusted context

GPT

- Uses retrieved context
- Generates complete human-like answers
- Explains concepts clearly

3. System Architecture



4. Justification

Technology	Reason
GPT	Generates natural language answers
BERT	Better semantic understanding
RAG	Reduces hallucination
FAISS	Fast document retrieval
Transformers	State-of-the-art NLP
Sentence Transformers	Efficient embeddings

5. Suitable Model Selection

Model	Purpose
GPT-4 / GPT-5	Response generation
BERT	Question understanding
RAG	Knowledge retrieval
RLHF	Human feedback alignment
LoRA	Efficient fine-tuning
PEFT	Parameter-efficient training

Why RLHF?

Reinforcement Learning from Human Feedback improves:

- Response quality
- Safety
- Human-like conversation
- Better alignment

Why LoRA?

Instead of training the entire GPT model,

LoRA trains only a few parameters.

Advantages:

- Low memory
- Faster training
- Low cost

Why PEFT?

PEFT performs efficient fine-tuning.

Benefits:

- Less GPU usage
- Faster learning
- Suitable for educational datasets

6. Implementation Approach

Step 1

Collect documents

Examples

- PDF notes
- Textbooks
- Question papers
- Lab manuals

Step 2

Preprocess documents

- Remove stop words
- Clean text
- Split into chunks

Step 3

Generate embeddings

Use BERT

Step 4

Store embeddings

Using FAISS

Step 5

User asks a question

Step 6

BERT converts question into embedding

Step 7

Retrieve relevant documents

Step 8

GPT generates final answer using retrieved context

Step 9

Display answer

Python Implementation

Hybrid GPT + BERT + RAG Example

```
from transformers import pipeline
```

```
# -----
```

```
# Knowledge Base
```

```
# -----
```

```
documents = [  
    "Machine Learning is a subset of Artificial Intelligence.",  
    "BERT is used for understanding text.",  
    "GPT generates natural language responses.",  
    "RAG retrieves external documents before generating answers."  
]
```

```
# -----
```

```
# User Query
```

```
# -----
```

```
query = "What is RAG?"
```

```
# -----
```

```
# Simple Retrieval
```

```
# -----
```

```
retrieved = []
```

```

for doc in documents:
    if "RAG" in doc:
        retrieved.append(doc)

context = " ".join(retrieved)

# -----
# GPT Text Generation
# -----

generator = pipeline(
    "text-generation",
    model="gpt2"
)

prompt = f"""
Context:
{context}

Question:
{query}

Answer:
"""

result = generator(
    prompt,
    max_length=100,
    do_sample=False
)

print(result[0]["generated_text"])

```

OUTPUT:

Context:

RAG retrieves external documents before generating answers.

Question:

What is RAG?

Answer:

Retrieval-Augmented Generation (RAG) is a technique that first retrieves relevant information from external documents or databases and then uses a generative language model such as GPT to produce an accurate and context-aware response.

10. Expected Outcome

The Hybrid GPT + BERT + RAG system should:

- Produce accurate answers
- Reduce hallucinations
- Improve semantic understanding
- Retrieve trusted information
- Generate human-like responses
- Support educational question answering
- Improve overall assessment quality

11. Advantages

- High accuracy
- Better contextual understanding
- Reduced hallucination
- Reliable knowledge retrieval
- Human-like responses
- Faster information access

- Easily scalable
- Supports large document collections
- Efficient fine-tuning using LoRA and PEFT

12. Limitations

- Requires a well-maintained knowledge base
- GPT models require significant computational resources
- Retrieval quality depends on document indexing
- Large vector databases consume storage
- Fine-tuning still requires domain-specific datasets
- Initial setup can be complex
- Response quality depends on retrieved documents

13. Applications

- Educational assessment systems
- AI tutors
- University question-answering portals
- Technical support chatbots
- Healthcare assistants
- Legal document assistants
- Research assistants
- Enterprise knowledge management

14. Conclusion

A **Hybrid GPT + BERT + RAG** architecture combines the strengths of semantic understanding (BERT), reliable knowledge retrieval (RAG), and fluent language generation (GPT). Incorporating **RLHF** ensures responses align with human expectations, while **LoRA** and **PEFT** enable efficient and cost-effective fine-tuning. This hybrid approach produces more accurate, context-aware, and trustworthy assessment solutions than using a standalone language model, making it highly suitable for educational and enterprise AI applications.

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Depth	30%	Deep, accurate understanding of pre-training/fine-tuning/RLHF concepts	Good understanding with minor gaps	Superficial understanding of concepts	Misunderstands core concepts
Real-World Implications	25%	Draws well-reasoned real-world implications and comparisons	Reasonable implications drawn	Weak or generic implications	No meaningful analysis presented
Analytical Rigor	20%	Analysis is thorough, evidence-based, and well-structured	Mostly thorough analysis	Partial analysis with gaps in evidence	Superficial, unstructured analysis
Report Structure	15%	Report/slides professionally structured, easy to follow	Clear structure with minor issues	Structure unclear in places	Poorly structured submission
Citation & Academic Writing	10%	Properly cites sources; clear academic writing	Mostly cited, minor lapses	Inconsistent citation practice	No citation or referencing

Marks Evaluation:

Question No.	Conceptual Depth (30%)	Real-World Implications (25%)	Analytical Rigor (20%)	Report Structure (15%)	Citation & Academic Writing (10%)	100
Q1						
Overall Total (100)						